

Dany Binu Luke

Roll No.: U23CS008
B.Tech – Computer Science and Engineering
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GitHub | Portfolio | LinkedIn
LeetCode | Codeforces

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. (CSE)	S.V. National Institute of Technology	7.22 (Current)	2023–Present
Senior Secondary	Central Board of Secondary Education	79%	2023
Secondary	Central Board of Secondary Education	90%	2021

PROJECTS

- MicroState – Real-Time Market Microstructure Analytics** 2026
Python, FastAPI, Next.js, Supabase, Docker Live Demo
 - Engineered a real-time system reconstructing Level-2 limit order books with event-driven microstructure analysis achieving 1-second latency on cryptocurrency data streams; implemented multi-timescale ML architecture with **92% prediction accuracy**.
 - Created full-stack dashboard with React and FastAPI supporting 100+ concurrent users via WebSocket streaming with real-time visualization of 15+ market metrics; containerized with Docker handling **50+ requests/second**.
- Fact-Checking & Contradiction Detection RAG System** 2026
FastAPI, Gemini API, OpenRouter, Ollama, Pinecone, Neo4j, Python Live Demo
 - Built a **RAG-based fact verification system** capable of extracting **5–10 factual claims per document**, performing semantic retrieval and classifying **SUPPORT/CONTRADICTION/NEUTRAL** relationships with **87% contradiction detection accuracy**.
 - Engineered a multi-provider AI inference pipeline using **Gemini, OpenRouter**, and local **Ollama Qwen3** fallback support; integrated **Pinecone** vector search with sub-**600 ms** retrieval latency and **Neo4j** graph visualization to map semantic relationships supporting **100+ interconnected claim relationships**.
- Ordinus Bot – Gesture-Controlled Pick and Place Robot** 2025
Python, MediaPipe, OpenCV, PySimpleGUI, Arduino GitHub
 - Engineered real-time gesture pipeline using **MediaPipe Hands** to extract 21 hand landmarks per frame; classified 6 distinct gestures with **90% accuracy**, mapping landmark coordinates to servo angles via a custom **OpenCV** preprocessing stage .
 - Built a **PySimpleGUI** dual-mode control interface with 6 live servo sliders, teaching-and-playback mode, JSON pose import/export, and a full event log; serial command dispatch over USB achieved **160–180 ms** end-to-end latency and **88%** pick-and-place task success.
- SentiStock AI – Stock Sentiment Analysis and Prediction** 2025
Python, NLP, Machine Learning, Transformers Live Demo
 - Constructed machine learning pipeline processing 500+ financial news articles daily using transformer-based sentiment analysis achieving **87% classification accuracy** for automated market trend prediction.
 - Implemented real-time data ingestion forecasting price movements with **78% directional accuracy**, delivering actionable insights faster than manual analysis approaches.

SKILLS

- Programming Languages:** Python, C, C++, Java, JavaScript, MySQL
- Tools / Frameworks:** PyTorch, OpenCV, MediaPipe, HuggingFace Transformers, NumPy, Pandas, MERN Stack, NextJS, FastAPI, Docker, Kubernetes, REST APIs, Framer Motion, Retrieval-Augmented Generation (RAG), Vector Databases
- Additional Skills:** Teamwork and Project Management, Effective written and verbal communication

KEY COURSES TAKEN

- Computer Science:** Data Structures, Algorithms, Database Management Systems, Computer Networks, Operating Systems, Object Oriented Programming, Compiler Design, Machine Learning, Artificial Intelligence, Social Network Analysis
- Mathematics:** Linear Algebra, Discrete Mathematics, Probability & Statistics
- Business:** Professional Ethics, Economics and Business Management(PEEBM), Innovation, Incubation and Entrepreneurship(IIE)

ACHIEVEMENTS

- Runner-Up in Genesis 2026** organized by Cutting Edge Visionaries (CEV), SVNIT.
- In Gold League** during the Google Cloud Study Jams 2024 & 2025.
- Among the Top 5%** in NPTEL course – Privacy and Security in Online Social Media.